Rough notes on:

Little-Parks effect

by Francisco Lobo

These notes were initiated with the purpose of gaining a deeper understanding of the Little-Parks effect, the main focus of my paper "Little-Parks effect in thin superconducting rings in diffusive and ballistic limits."

THE LITTLE–PARKS EFFECT

To better understand the Little–Parks effect, one must first return to the framework of Ginzburg–Landau (GL) theory. GL theory is a functional minimization theory that describes a phenomenological free energy in terms of a complex order parameter Δ near the superconducting–normal phase transition, i.e. in the vicinity of the critical temperature.

Let us consider the system depicted in Fig. 1, consisting of a thin superconducting ring whose thickness d is much smaller than the coherence length ξ, the characteristic length scale over which the order parameter varies appreciably (or, in a BCS framework, the effective size of a Cooper pair), and also much smaller than the magnetic penetration depth λ, which characterizes the exponential decay of a magnetic field inside a superconductor.

Under these conditions, the magnitude of the order parameter cannot vary significantly in space, since any such variation on these length scales would incur a large energetic cost compared to the condensation energy and therefore would not correspond to the ground state. Moreover, given the constant magnitude of the order parameter, its phase must be single-valued: if it winds once around the ring and returns to the same point, the total phase winding must be an integer multiple of 2π. The appropriate ansatz for the order parameter therefore reads

$$\Delta(\boldsymbol{r}) = \Delta_0 e^{in\phi}. \tag{1}$$

Given the geometry of the system, the total magnetic flux threading the area enclosed by an arbitrary closed path C around the ring is the sum of the externally applied flux and the flux generated by the supercurrents circulating within the ring. The central idea underlying the Little–Parks effect is that this total flux is restricted to integer multiples of the flux quantum Φ_0. This fluxoid quantization follows directly from the requirement that the superconducting order

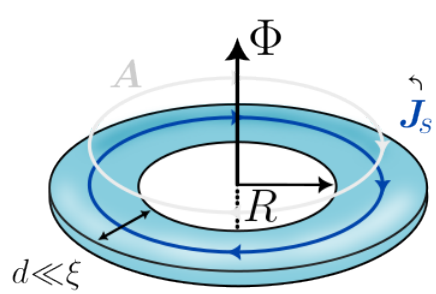

Figure 1. Thin superconducting ring

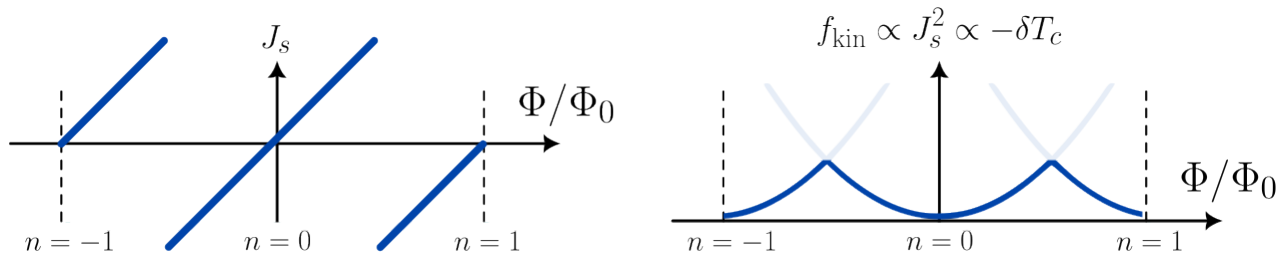

Figure 2. Little-Parks oscillations

parameter be single-valued. Explicitly, one may derive

$$\Phi_{\rm total} = \oint_C \boldsymbol{A}(\boldsymbol{r}) \cdot d\boldsymbol{\ell} + \frac{m^* c}{|\Delta|^2 e^{*2}} \oint_C \boldsymbol{J}_s \cdot d\boldsymbol{\ell} \tag{2}$$

$$= \oint_C \boldsymbol{A}(\boldsymbol{r}) \cdot d\boldsymbol{\ell} + \frac{m^* c}{e^*} \oint_C \boldsymbol{v}_s \cdot d\boldsymbol{\ell} \tag{3}$$

$$= \oint_C \boldsymbol{A}(\boldsymbol{r}) \cdot d\boldsymbol{\ell} + \frac{m^* c}{e^*} \oint_C \left(\frac{\hbar}{m^*} \boldsymbol{\nabla}\varphi - \frac{e^*}{m^* c} \boldsymbol{A} \right) \cdot d\boldsymbol{\ell} \tag{4}$$

$$= \frac{\hbar c}{e^*} \oint_C \boldsymbol{\nabla}\varphi \cdot d\boldsymbol{\ell} \tag{5}$$

$$= \frac{\hbar c}{e^*} 2\pi n \tag{6}$$

$$= n \frac{hc}{e^*} = n\Phi_0. \tag{7}$$

In an analogous derivation, one finds that the circulating supercurrent is proportional to $(n - \Phi/\Phi_0)$, while the associated kinetic energy E_k is proportional to $(n - \Phi/\Phi_0)^2$. Each integer value of n corresponds to a distinct parabolic free-energy branch, with adjacent parabolas intersecting at half-integer values of the flux. As the applied flux is increased past such a half-integer crossing, the global free-energy minimum shifts from the n-th parabola to the $(n+1)$-th parabola. This transition is accompanied by a reversal of the circulating supercurrent. The resulting periodic dependence of the free energy, and consequently of the critical temperature T_c, on the applied magnetic flux modulo Φ_0 constitutes the Little–Parks effect.

Further insight may be gained from the perspective of Ginzburg–Landau functional minimization. Near the superconducting–normal phase transition, the free energy must vary continuously, so that

$$f_{\rm SC} - f_{\rm N} = f_{\rm condensate} + f_{\rm kinetic} + f_{\rm magnetic}. \tag{8}$$

For a thin-walled ring with $d \ll \lambda$, magnetic screening is negligible and the magnetic field inside the superconductor is essentially equal to the applied field. As a consequence, the magnetic free-energy contribution is the same in the normal and superconducting states and cancels in the free-energy difference. The remaining contribution may then be written as

$$\delta f = (a + E_k)|\Delta|^2 + b|\Delta|^4, \tag{9}$$

where $a = \alpha(T - T_{c0})$. The superconducting instability occurs when the coefficient of $|\Delta|^2$ changes sign. The positive-definite kinetic energy E_k therefore shifts the transition to lower temperatures according to

$$T_c(\Phi) = T_{c0} - \frac{E_k(\Phi)}{\alpha}. \tag{10}$$

Thus, the flux-induced superfluid kinetic energy raises the free-energy cost of condensation and leads directly to a periodic suppression of the critical temperature.

Finally, it is essential that the ring be thin-walled for this simplified description to apply. In thick-walled rings, spatial variations of the order parameter, magnetic field inhomogeneity, vortex penetration, multiple fluxoid states across the thickness, and surface superconductivity must be taken into account, leading to a substantially more complex behavior. Further discussion of these effects may be found in `https://arxiv.org/pdf/cond-mat/9907341`.